



1 · Synthetic factory data

engine/factory-data.js

3 production lines · 10 client orders · line–order compatibilities · cost coefficients

Reproducible incident: Line 2 stops 10:00 → 14:00 (4 h)

Illustrative synthetic data — no customer data

fresh scenario object

2 · Deterministic twin engine

engine/twin-engine.js

Validate scenario → Analyze incident → Generate 3 constrained plans → Score KPIs → Approval event

Hard constraints on every candidate: line compatibility, no double-booking, incident window, deadlines

Explainable search-and-planning AI — deterministic · no LLM · no external model · no training data

three comparable recovery plans + before/after KPIs

3 · Decision interface (production manager)

app/index.html · app.js · styles.css

Nominal 10/10 → Incident → Compare strategies → Gantt preview → Approve → Audit trail → Reset

Side-by-side Service / Cost / Stability trade-offs, each previewed on a revised Gantt before approval

scripts/serve.mjs → http://127.0.0.1:4173 · Node.js 22 · zero dependencies · zero external requests

explicit approval only after human review

4 · Human authority & audit

human-in-command

The production manager approves or rejects; every approval becomes a timestamped audit event.

It never sends a command to a machine — it informs the decision; it does not replace the decider.

Independent exact verifier

engine/exact-benchmark.js

17,856 candidate schedules

exhaustive bounded enumeration

10,440 feasible schedules

three policy objectives

3 unique optima

Service · Cost · Stability — each verified and unique

Kept separate from the live planner so it can detect — rather than reproduce — a planning error.

optima cross-checked

Verification & evidence

- npm run check — 9/9 acceptance & regression checks
tests/twin-engine.test.mjs
- npm run benchmark:exact — re-proves the three optima
instant re-verification during jury Q&A
- Real-browser replay evidence (Chrome DevTools Protocol)
screenshots · machine-read UI metrics · zero-network trace
- SHA-256-frozen deliverables & immutable git tags
phase2-final · phase2-submission-final

Deployment boundary

Fully offline · zero external dependencies · read-only by design · no PLC / machine / MES / ERP / cloud access

All cost figures are illustrative and synthetic — never a real ROI claim.